

EVE-NG MULTI-VENDOR LABORATORY

Standard Operating Procedure — Multi-Vendor Firewall HA, IPsec VPN and BGP Overlay

Palo Alto Active/Passive HA · FortiGate · Juniper vSRX · Huawei CE12800

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1. Purpose

This SOP defines the procedure to build, verify and test a multi-vendor firewall environment comprising a Palo Alto Networks Active/Passive high-availability cluster, route-based IPsec VPN tunnels to a FortiGate and a Juniper vSRX firewall, an eBGP overlay carrying site LAN prefixes across those tunnels, and two Huawei CE12800 switches providing Layer 2 transit and LAN distribution.

The build procedure in Section 6 contains the configuration exported from each device, so the environment can be rebuilt from an empty EVE-NG topology. Section 7 covers read-only verification, Section 8 covers service-impacting testing, and Sections 9 to 12 record the validation outcome.

1.1 Outstanding Items Before Final Sign-Off

#	Item	Reference
1	FortiGate VPN1 tunnel interface has no IP address, so the BGP session to the Palo Alto cluster cannot establish	Section 10.1
2	Static route test for 192.168.20.0/24 via tunnel.1 currently masks the BGP failure and should be removed or reclassified once BGP is restored	Section 10.2
3	Palo Alto passive link state is configured as auto ; the design intent is shutdown	Section 6.3
4	BGP-learned prefix 192.168.30.0/24 appears in the local RIB and FIB but not in the captured route table output; re-capture in a single session	Section 10.3
5	Palo Alto BGP export policy lan is scoped to the FGT-Peer group only	Section 10.4
6	Palo Alto rulebase contains an any-any allow rule above the zone-specific rules	Section 10.5
7	End-to-end ping and traceroute between PC1, PC2 and PC3, and packet-loss measurement during failover, are not yet captured	Section 8.4
8	IKE and IPsec proposals use DES and must be replaced before any production adaptation	Section 11.7

2. Scope

2.1 In Scope

- Construction of the EVE-NG topology and the link plan between all nodes.
- Layer 2 configuration and verification on both Huawei CE12800 switches.
- Palo Alto Active/Passive HA build, configuration synchronisation and link monitoring.
- Palo Alto, FortiGate and Juniper vSRX interface, zone, policy and routing configuration.
- Route-based IKEv2 IPsec tunnel build between the Palo Alto cluster and both peer firewalls.
- eBGP overlay configuration across the tunnels.
- Read-only verification of HA, IPsec, BGP and routing state on all platforms.
- Controlled HA failover testing, including preemption behaviour.

2.2 Out of Scope

- EVE-NG installation, image upload, licensing and node sizing.
- Threat prevention, URL filtering, NAT and production-grade policy design.
- Production change control, maintenance windows and approval workflow.
- Performance, throughput and scale testing.
- Backup and monitoring platform integration.

3. Lab Overview

3.1 Laboratory Summary

The environment is hosted in EVE-NG and models a small enterprise with a data centre and two remote sites. The data centre firewall function is delivered by a Palo Alto HA pair. Each remote site is protected by a different vendor's firewall, which makes the environment suitable for validating multi-vendor IPsec and BGP interoperability.

3.2 Device List

Device	Platform and software	Role	Configuration file
PA-FW-01	Palo Alto PA-VM, PAN-OS 8.1.0, content 769-4439	HA primary, data centre firewall	configs/PA-FW-01.txt
PA-FW-02	Palo Alto PA-VM, PAN-OS 8.1.0, content 769-4439	HA secondary, data centre firewall	configs/PA-FW-02.txt
FortiGate	FortiGate VM, FortiOS 7.6.2 build 3462	Remote site edge firewall	configs/Fortinet.txt
vSRX-NG	Juniper vSRX, Junos 24.4R1.9	Remote site edge firewall	configs/srx.txt
Core-SW-1	Huawei CE12800, VRP V200R005C10SPC607B607, sysname switch A	Layer 2 transit switch	configs/Core-SW-01.txt
Core-SW-2	Huawei CE12800, VRP V200R005C10SPC607B607, sysname Core-SW-2	LAN distribution and transit switch	configs/Core-SW-02.txt
PC1	Test host	Remote site LAN client, 192.168.20.0/24	—
PC2	Test host	Data centre LAN client, 192.168.10.0/24	—
PC3	Test host	Remote site LAN client, 192.168.30.0/24	—

Both Huawei switches use a CE-MPUB main processing board and a CE-LPUE line card.

3.3 High-Level Topology Description

The FortiGate connects to Core-SW-1, which provides Layer 2 transit in VLAN 20 to both Palo Alto units. The Palo Alto pair connects to Core-SW-2 for the data centre LAN in VLAN 10 and for transit to the Juniper vSRX in VLAN 30. The vSRX serves its own site LAN. The two Palo Alto units are directly cabled to each other for the HA1 control link and the HA2 data link. All firewalls have a dedicated out-of-band management interface.

Because the cluster is Active/Passive, only the active unit forwards data traffic at any time. IPsec tunnels and BGP sessions therefore terminate on the cluster as a whole rather than on an individual unit, and are preserved across a failover by state synchronisation.

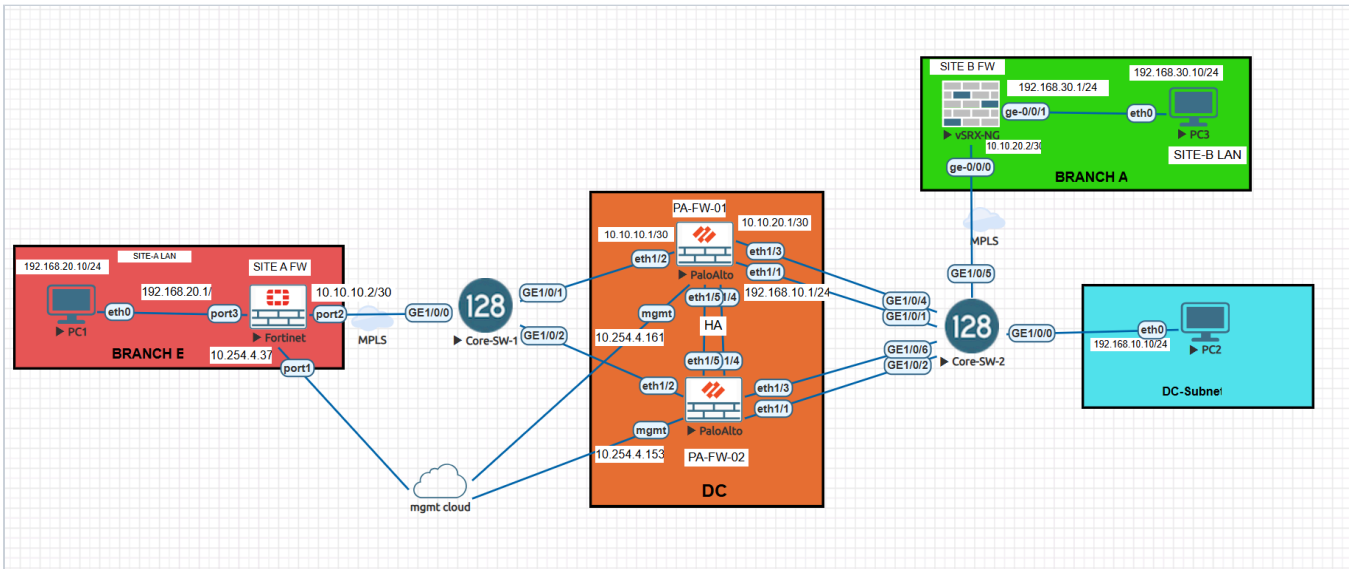


Figure 1 — Multi-vendor laboratory topology, including transit subnets, management network and site LANs.

4. Network Design and Addressing

4.1 Management Addresses

Device	Management interface	Addressing
PA-FW-01	mgmt	10.254.4.161/24
PA-FW-02	mgmt	10.254.4.153/24
FortiGate	port1	DHCP, allowaccess ping/https/ssh/http
vSRX	fxp0.0	No address configured; HTTP and HTTPS management enabled on fxp0.0
Core-SW-1	MEth0/0/0	No address configured; console and SSH only
Core-SW-2	MEth0/0/0	No address configured; console and SSH only

4.2 LAN Subnets

Site	Subnet	Gateway interface	Test host
Data centre LAN	192.168.10.0/24	Palo Alto ethernet1/1, 192.168.10.1	PC2, 192.168.10.10
FortiGate site LAN	192.168.20.0/24	FortiGate port3, 192.168.20.1	PC1
vSRX site LAN	192.168.30.0/24	vSRX ge-0/0/1.0, 192.168.30.1	PC3

4.3 Transit Subnets and VLANs

Link	Subnet	Switch and VLAN	Endpoints
Palo Alto to FortiGate	10.10.10.0/30	Core-SW-1, VLAN 20 (FA-PA)	Palo Alto ethernet1/2 .1, FortiGate port2 .2
Palo Alto to vSRX	10.10.20.0/30	Core-SW-2, VLAN 30 (pa-srx)	Palo Alto ethernet1/3 .1, vSRX ge-0/0/0.0 .2
Data centre LAN	192.168.10.0/24	Core-SW-2, VLAN 10 (pa-lan)	Palo Alto ethernet1/1, PC2

4.4 HA Links

Link	Interface	PA-FW-01	PA-FW-02	Mask
HA1 control	ethernet1/4	10.1.1.1	10.1.1.2	255.255.255.252
HA2 data	ethernet1/5	10.2.2.1	10.2.2.2	255.255.255.252

Both HA interfaces are set to interface type `ha` with `link-state auto`. No HA1 backup link is configured.

4.5 Tunnel Interfaces

Device	Tunnel interface	Address	IPsec tunnel / gateway	Peer
PA-FW-01	tunnel.1	172.24.10.1/30	Tunnel <code>VPN1</code> , gateway <code>VPN1</code>	FortiGate, 10.10.10.2
PA-FW-01	tunnel.2	172.16.1.1/30	Tunnel <code>SRXVPN</code> , gateway <code>VPN2</code>	vSRX, 10.10.20.2
vSRX	st0.0	172.16.1.2/30	VPN <code>pa-vpn</code> , gateway <code>pa-gw</code>	Palo Alto, 10.10.20.1
FortiGate	VPN1	Not configured	Phase 1 <code>VPN1</code> , phase 2 <code>PA</code>	Palo Alto, 10.10.10.1

Both Palo Alto tunnel interfaces are in security zone `VPN`, in virtual router `default`, and use the `Test` interface management profile. The FortiGate tunnel interface is bound to port2 but has no IP address, which is the defect recorded in Section 10.1.

4.6 BGP Autonomous Systems

Device	AS number	Router ID	Notes
PA-FW-01 / PA-FW-02	65001	1.1.1.1	<code>install-route yes</code> , graceful restart enabled, dampening profile <code>default</code>
FortiGate	65000	2.2.2.2	eBGP multipath, additional-path and graceful restart enabled
vSRX	65003	Derived from interface addressing	Group <code>overlay</code> , type external

4.7 BGP Peers

Local device	Peer group	Peer name	Local address	Peer address	Remote AS	Validated state
PA-FW-01	<code>FGT-Peer</code>	<code>fgt-peer</code>	172.24.10.1/30, tunnel.1	172.24.10.2	65000	Connect, not established
PA-FW-01	<code>srx</code>	<code>SRX-peer</code>	172.16.1.1/30, tunnel.2	172.16.1.2	65003	Established
FortiGate	—	172.24.10.1	Update source <code>VPN1</code>	172.24.10.1	65001	Active, never established
vSRX	<code>overlay</code>	172.16.1.1	st0.0, 172.16.1.2	172.16.1.1	65001	Established

Palo Alto peer connection options on both peer groups: hold time 90 seconds, keep-alive 30 seconds, idle hold time 15 seconds, minimum route advertisement interval 30 seconds, multihop 0, maximum prefixes 5000, sender-side loop detection enabled, next hop set to self on export.

FortiGate neighbour options: advertisement interval 10 seconds, connect timer 10 seconds, `ebgp-enforce-multihop`, `next-hop-self`, `soft-reconfiguration`, dynamic capability and graceful restart capability enabled.

4.8 Route Advertisement Plan

Prefix	Origin	Injection method	Validated outcome
192.168.10.0/24	PA-FW-01	Redistribution profile <code>LAN</code> (connected, ethernet1/1) with BGP redistribution rule <code>LAN</code> , origin incomplete; export policy rule <code>lan</code> used by <code>FGT-Peer</code>	Received by the vSRX with AS path <code>65001 ?</code> ; not received by the FortiGate because the session is down
192.168.20.0/24	FortiGate	<code>config router bgp network statement</code>	Present in the FortiGate BGP table only; not advertised because the session is down
192.168.30.0/24	vSRX	Export policy <code>bgp-export</code> , term 1, protocol direct, route-filter 192.168.30.0/24 exact	Received by PA-FW-01 via 172.16.1.2

PA-FW-01 also carries static route `test` for 192.168.20.0/24 via tunnel.1 with metric 10. See Section 10.2.

5. Prerequisites

5.1 Required Access

Requirement	Detail
Palo Alto	Administrator access by SSH or console, plus web UI, on both units
FortiGate	Administrator access by CLI console and web UI
Juniper vSRX	CLI access with <code>root</code> privileges
Huawei CE12800	CLI access, VRP v8, console or SSH
Test hosts	Terminal access on PC1, PC2 and PC3
EVE-NG	Lab owner or administrator rights to create and edit the topology

5.2 Required Backups

A configuration backup must exist for every device before any change is applied:

- Palo Alto: exported named configuration snapshot from each unit.
- FortiGate: configuration backup file.
- Juniper vSRX: saved configuration, and a rescue configuration where available.
- Huawei: saved running configuration on both switches.

5.3 Required Software and Image State

- Both Palo Alto units must run identical software and content versions. Validation confirmed Build Release 8.1.0 and Application Content 769-4439 on both peers, with all version compatibility fields reporting `Match`.
- FortiOS 7.6.2 build 3462, Junos 24.4R1.9 and VRP V200R005C10SPC607B607 were used for this build.
- Images must be present in EVE-NG and correctly permissioned before the lab is started.

5.4 Required Pre-Shared Keys

Each IPsec gateway requires a pre-shared key agreed between the two peers. All three firewalls use IKEv2 with pre-shared key authentication. The Palo Alto uses the same key on gateways `VPN1` and `VPN2`. Key values are stored encrypted in the device configurations and are redacted from this document.

5.5 Required Management Reachability

Confirm that each management address responds and that administrative sessions can be opened to every device before starting. The Palo Alto units are reachable on 10.254.4.161 and 10.254.4.153. The FortiGate

obtains its management address by DHCP on port1. The vSRX and both switches are reached over console in this build.

5.6 Required Console or CLI Access

Console access must remain available throughout the build. HA changes, tunnel changes and routing changes can all interrupt in-band management, so an out-of-band path is required to recover a device if a change is applied incorrectly.

6. Build Procedure

Each subsection below contains the configuration as exported from the running devices. Apply the commands in the order given: switching first, then the HA cluster, then interfaces and zones, then tunnels, then BGP.

6.1 Build the EVE-NG Topology

Step 1 — Add the nodes. Create a new lab and add two Palo Alto PA-VM instances (PA-FW-01 and PA-FW-02), one FortiGate, one Juniper vSRX, two Huawei CE12800 switches, three test hosts, and one management cloud object bridged to the out-of-band management network.

Step 2 — Cable the links according to the following plan:

From	Interface	To	Interface	Purpose
FortiGate	port1	Management cloud	—	Out-of-band management, DHCP
FortiGate	port2	Core-SW-1	GE1/0/0	Transit to the Palo Alto cluster, VLAN 20
FortiGate	port3	PC1	eth0	Site LAN 192.168.20.0/24
Core-SW-1	GE1/0/1	PA-FW-01	ethernet1/2	Transit, VLAN 20
Core-SW-1	GE1/0/2	PA-FW-02	ethernet1/2	Transit, VLAN 20
PA-FW-01	ethernet1/1	Core-SW-2	GE1/0/1	Data centre LAN, VLAN 10
PA-FW-02	ethernet1/1	Core-SW-2	GE1/0/2	Data centre LAN, VLAN 10
PA-FW-01	ethernet1/3	Core-SW-2	GE1/0/4	Transit to the vSRX, VLAN 30
PA-FW-02	ethernet1/3	Core-SW-2	GE1/0/6	Transit to the vSRX, VLAN 30
PA-FW-01	ethernet1/4	PA-FW-02	ethernet1/4	HA1 control link, direct
PA-FW-01	ethernet1/5	PA-FW-02	ethernet1/5	HA2 data link, direct
Core-SW-2	GE1/0/0	PC2	eth0	Data centre LAN host, VLAN 10
Core-SW-2	GE1/0/5	vSRX	ge-0/0/0	Transit to the Palo Alto cluster, VLAN 30
vSRX	ge-0/0/1	PC3	eth0	Site LAN 192.168.30.0/24
PA-FW-01 / PA-FW-02	mgmt	Management cloud	—	Out-of-band management

Step 3 — Start the nodes in this order: switches, firewalls, then test hosts, so that Layer 2 transit is available before the firewalls attempt to form adjacencies.

6.2 Configure Huawei CE12800 Switching

Both switches provide Layer 2 transit only. No Layer 3 interfaces or routing are configured.

Core-SW-1 — transit switch. VLAN 20 carries the 10.10.10.0/30 transit segment between the FortiGate and both Palo Alto units. STP is disabled on the three firewall-facing ports, since they cannot form a loop and do not participate in spanning tree.

```
system-view
sysname switch A
#
vlan batch 20
#
vlan 20
description FA-PA
#
interface GE1/0/0
port link-type access
port default vlan 20
stp disable
#
interface GE1/0/1
port link-type access
port default vlan 20
stp disable
#
interface GE1/0/2
port link-type access
port default vlan 20
stp disable
#
interface GE1/0/3
port link-type access
port default vlan 20
#
quit
return
save
```

Core-SW-2 — LAN and transit switch. VLAN 10 carries the data centre LAN and VLAN 30 carries the 10.10.20.0/30 transit segment to the vSRX, keeping the two subnets in separate broadcast domains.

```
system-view
sysname Core-SW-2
#
vlan batch 10 30
#
vlan 10
description pa-lan
#
vlan 30
description pa-srx
#
interface GE1/0/0
port link-type access
port default vlan 10
stp disable
#
interface GE1/0/1
port link-type access
port default vlan 10
stp disable
#
interface GE1/0/2
port link-type access
port default vlan 10
stp disable
#
interface GE1/0/4
port link-type access
port default vlan 30
stp disable
#
interface GE1/0/5
port link-type access
port default vlan 30
stp disable
#
interface GE1/0/6
port link-type access
port default vlan 30
stp disable
#
quit
return
save
```

Verify before continuing.

```
display vlan
display interface brief
display mac-address
```

```

<switch A>display vlan
The total number of vlans is : 2
-----
U: Up;           D: Down;           TG: Tagged;       UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
MAC-LRN: MAC-address learning; STAT: Statistic;
BC: Broadcast; MC: Multicast; UC: Unknown-unicast;
FWD: Forward; DSD: Discard;
-----

VID          Ports
-----
1            UT:GE1/0/4(U)    GE1/0/5(U)    GE1/0/6(U)    GE1/0/7(U)
             GE1/0/8(U)    GE1/0/9(U)
20           UT:GE1/0/0(U)    GE1/0/1(U)    GE1/0/2(U)    GE1/0/3(U)

VID  Type      Status  Property  MAC-LRN  STAT   BC  MC  UC  Description
-----
1    common    enable  default   enable   disable FWD FWD FWD VLAN 0001
20   common    enable  default   enable   disable FWD FWD FWD FA-PA
<switch A>

```

Figure 2 — Core-SW-1 VLAN 20 assignment for the FortiGate to Palo Alto transit segment.

```

<switch A>display interface brief
PHY: Physical
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(b): BFD down
(e): ETHOAM down
(d): Dampening Suppressed
(p): port alarm down
(dl): DLDAP down
(c): CFM down
InUti/OutUti: input utility rate/output utility rate
Interface          PHY      Protocol  InUti  OutUti  inErrors  outErrors
GE1/0/0             up       up        0%    0.01%    0         0
GE1/0/1             up       up        0.01%  0%      0         0
GE1/0/2             up       up        0%    0%      0         0
GE1/0/3             up       up        0%    0%      0         0
GE1/0/4             up       up        0%    0%      0         0
GE1/0/5             up       up        0%    0%      0         0
GE1/0/6             up       up        0%    0%      0         0
GE1/0/7             up       up        0%    0%      0         0
GE1/0/8             up       up        0%    0%      0         0
GE1/0/9             up       up        0%    0%      0         0
MEth0/0/0           up       down      0%    0%      0         0
NULL0              up       up(s)     0%    0%      0         0
<switch A>

```

Figure 3 — Core-SW-1 interface status, confirming firewall-facing ports are up.

```

<switch A>display mac-address
Flags: * - Backup
BD   : bridge-domain   Age : dynamic MAC learned time in seconds
-----
MAC Address      VLAN/VSI/BD   Learned-From   Type           Age
-----
5000-0001-0001  20/-/-       GE1/0/0        dynamic        -
5000-0003-0002  20/-/-       GE1/0/1        dynamic        -
5000-0001-0001  20/-/-       GE1/0/0        dynamic        -
5000-0003-0002  20/-/-       GE1/0/1        dynamic        -
-----
Total items: 4
<switch A>

```

Figure 4 — Core-SW-1 MAC address table, confirming the attached firewall interfaces are learned.

```

Total Items: 0
<Core-SW-2>display vlan
The total number of vlans is : 3
-----
U: Up;           D: Down;           TG: Tagged;           UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
MAC-LRN: MAC-address learning; STAT: Statistic;
BC: Broadcast; MC: Multicast; UC: Unknown-unicast;
FWD: Forward; DSD: Discard;
-----

VID      Ports
-----
1        UT:GE1/0/3(U)   GE1/0/7(U)   GE1/0/8(U)   GE1/0/9(U)
10       UT:GE1/0/0(U)   GE1/0/1(U)   GE1/0/2(U)
30       UT:GE1/0/4(U)   GE1/0/5(U)   GE1/0/6(U)

VID  Type      Status  Property  MAC-LRN  STAT   BC  MC  UC  Description
-----
1  common  enable  default  enable   disable FWD FWD FWD VLAN 0001
10 common  enable  default  enable   disable FWD FWD FWD pa-lan
30 common  enable  default  enable   disable FWD FWD FWD pa-srx
<Core-SW-2>

```

Figure 5 — Core-SW-2 VLAN 10 and VLAN 30 assignment.

```

<Core-SW-2>display interface brief
PHY: Physical
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
(b): BFD down
(e): ETHOAM down
(d): Dampening Suppressed
(p): port alarm down
(dl): DLDLP down
(c): CFM down
InUti/OutUti: input utility rate/output utility rate
Interface          PHY      Protocol  InUti  OutUti  inErrors  outErrors
GE1/0/0             up       up        0%     0%      0         0
GE1/0/1             up       up        0%     0%      0         0
GE1/0/2             up       up        0%     0%      0         0
GE1/0/3             up       up        0%     0%      0         0
GE1/0/4             up       up        0%     0%      0         0
GE1/0/5             up       up        0%     0%      0         0
GE1/0/6             up       up        0%     0%      0         0
GE1/0/7             up       up        0%     0%      0         0
GE1/0/8             up       up        0%     0%      0         0
GE1/0/9             up       up        0%     0%      0         0
MEth0/0/0           up       down      0%     0%      0         0
NULL0              up       up(s)     0%     0%      0         0
<Core-SW-2>

```

Figure 6 — Core-SW-2 interface status.

```

<Core-SW-2>display mac-address
Flags: * - Backup
BD   : bridge-domain   Age : dynamic MAC learned time in seconds
-----
MAC Address      VLAN/VSI/BD   Learned-From   Type           Age
-----
5000-0003-0001  10/-/-       GE1/0/1        dynamic        -
5000-0004-0001  30/-/-       GE1/0/5        dynamic        -
5000-0003-0003  30/-/-       GE1/0/4        dynamic        -
5000-0003-0001  10/-/-       GE1/0/1        dynamic        -
5000-0004-0001  30/-/-       GE1/0/5        dynamic        -
5000-0003-0003  30/-/-       GE1/0/4        dynamic        -
-----
Total items: 6
<Core-SW-2>

```

Figure 7 — Core-SW-2 MAC address table.

6.3 Configure the Palo Alto HA Pair

Build the cluster before configuring tunnels or routing, so that all later configuration is synchronised to the peer automatically. Enter configuration mode with `configure` on each unit.

PA-FW-01 — device priority 50, preferred active.

```

set deviceconfig high-availability enabled yes
set deviceconfig high-availability group group-id 1
set deviceconfig high-availability group peer-ip 10.1.1.2
set deviceconfig high-availability group mode active-passive passive-link-state auto
set deviceconfig high-availability group election-option device-priority 50
set deviceconfig high-availability group election-option preemptive yes
set deviceconfig high-availability group election-option heartbeat-backup yes
set deviceconfig high-availability group election-option timers recommended
set deviceconfig high-availability group monitoring link-monitoring link-group test interface [
ethernet1/1 ethernet1/2 ethernet1/3 ethernet1/4 ethernet1/5 ]
set deviceconfig high-availability interface ha1 port ethernet1/4
set deviceconfig high-availability interface ha1 ip-address 10.1.1.1
set deviceconfig high-availability interface ha1 netmask 255.255.255.252
set deviceconfig high-availability interface ha2 port ethernet1/5
set deviceconfig high-availability interface ha2 ip-address 10.2.2.1
set deviceconfig high-availability interface ha2 netmask 255.255.255.252
set network interface ethernet ethernet1/4 ha
set network interface ethernet ethernet1/4 link-state auto
set network interface ethernet ethernet1/5 ha
set network interface ethernet ethernet1/5 link-state auto
commit

```

PA-FW-02 — device priority 100, standby.

```

set deviceconfig high-availability enabled yes
set deviceconfig high-availability group group-id 1
set deviceconfig high-availability group peer-ip 10.1.1.1
set deviceconfig high-availability group mode active-passive passive-link-state auto
set deviceconfig high-availability group election-option device-priority 100
set deviceconfig high-availability group election-option preemptive yes
set deviceconfig high-availability group election-option heartbeat-backup yes
set deviceconfig high-availability group election-option timers recommended
set deviceconfig high-availability group monitoring link-monitoring link-group test interface [
ethernet1/1 ethernet1/2 ethernet1/3 ethernet1/4 ethernet1/5 ]
set deviceconfig high-availability interface ha1 port ethernet1/4
set deviceconfig high-availability interface ha1 ip-address 10.1.1.2
set deviceconfig high-availability interface ha1 netmask 255.255.255.252
set deviceconfig high-availability interface ha2 port ethernet1/5
set deviceconfig high-availability interface ha2 ip-address 10.2.2.2
set deviceconfig high-availability interface ha2 netmask 255.255.255.252
commit

```

Design notes.

- In PAN-OS the lower device priority value is preferred, so PA-FW-01 at 50 becomes active.
- `timers recommended` applies the platform defaults observed during validation: hello interval 8000 ms, heartbeat ping interval 2000 ms, promotion hold 2000 ms, preemption hold 1 minute, monitor fail hold down 1 minute, maximum flaps 3.
- The link monitoring group `test` uses failure condition `any`, so any one monitored interface going down triggers a failover.
- Passive link state is set to `auto`. The design intent is `shutdown`, which holds the passive unit's data ports down and prevents MAC and ARP churn on the transit switches. Change with `set deviceconfig high-availability group mode active-passive passive-link-state shutdown` on both units.

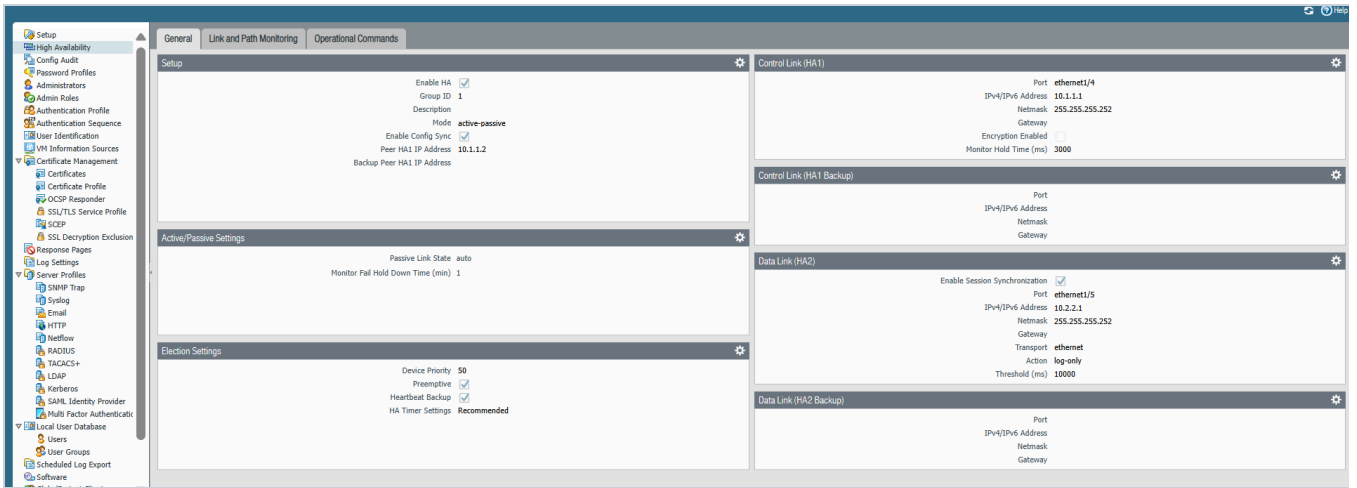


Figure 8 — PA-FW-01 HA configuration and status, showing the active role and synchronised configuration.

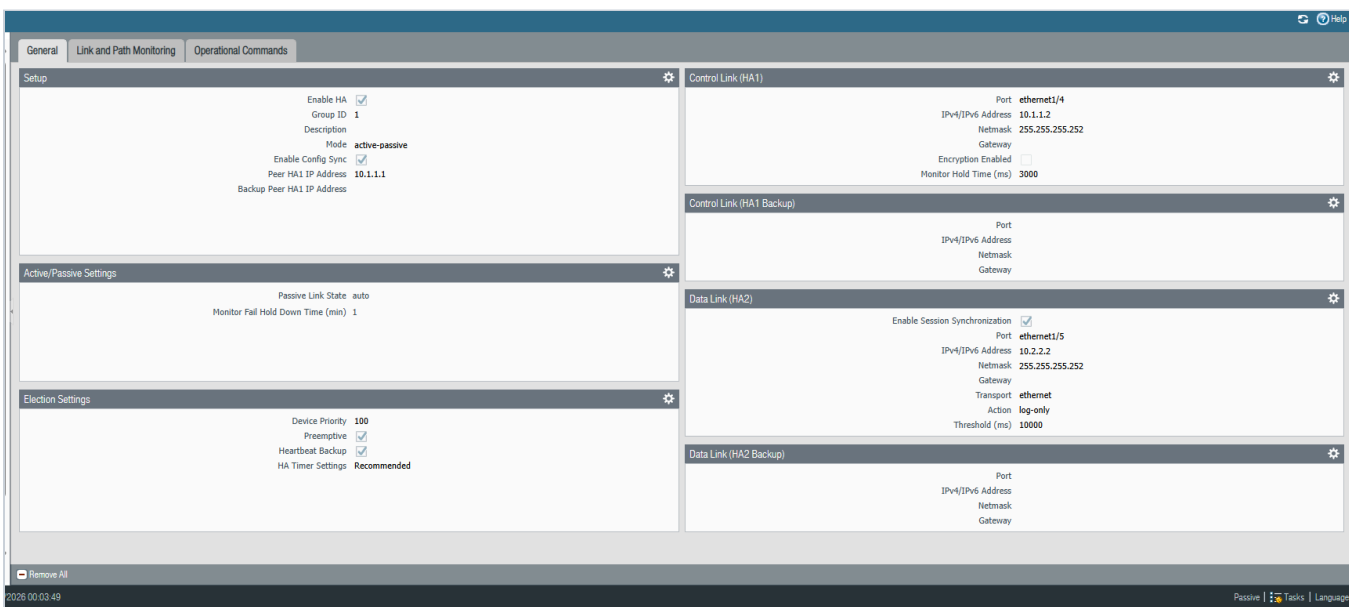


Figure 9 — PA-FW-02 HA configuration and status, showing the passive role.

General
Link and Path Monitoring
Operational Commands

Link Monitoring

Enabled ☒

Failure Condition any

Link Group

	Name	Enabled	Group Failure Condition	Interfaces
☐	test	☒	any	ethernet1/1 ethernet1/2 ethernet1/3 ethernet1/4 ethernet1/5

+ Add
- Delete

Path Monitoring

Enabled ☒

Failure Condition any

Path Group

	Name	Type	Enabled	Failure Condition	Source IP	Destination IP	Ping Interval	Ping Count
--	------	------	---------	-------------------	-----------	----------------	---------------	------------

+ Add Virtual Wire Path
+ Add VLAN Path
+ Add Virtual Router Path
- Delete

Figure 10 — PA-FW-01 link monitoring group `test` and monitored interface states.

General
Link and Path Monitoring
Operational Commands

Link Monitoring

Enabled ☒

Failure Condition any

Link Group

	Name	Enabled	Group Failure Condition	Interfaces
☐	test	☒	any	ethernet1/1 ethernet1/2 ethernet1/3 ethernet1/4 ethernet1/5

+ Add
- Delete

Path Monitoring

Enabled ☒

Failure Condition any

Path Group

	Name	Type	Enabled	Failure Condition	Source IP	Destination IP	Ping Interval	Ping Count
--	------	------	---------	-------------------	-----------	----------------	---------------	------------

+ Add Virtual Wire Path
+ Add VLAN Path
+ Add Virtual Router Path
- Delete

Figure 11 — PA-FW-02 link monitoring configuration.

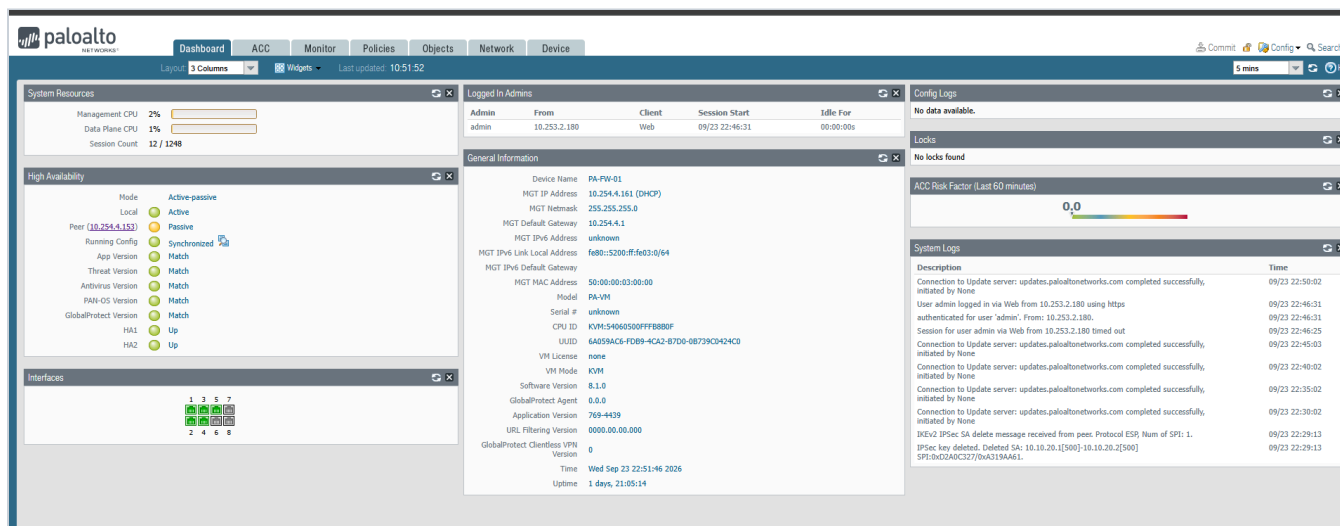


Figure 12 — PA-FW-01 dashboard summary, including the HA widget.

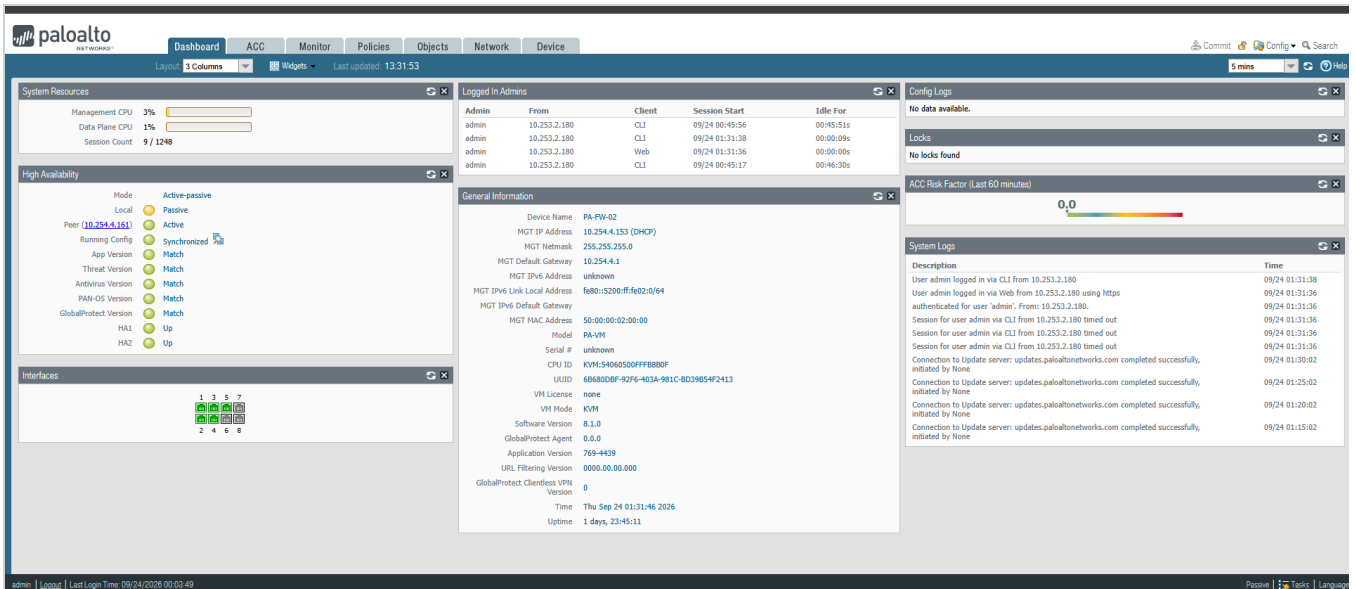


Figure 13 — PA-FW-02 dashboard summary.

6.4 Configure Palo Alto Interfaces, Zones and Virtual Router

Apply on the active unit; the configuration synchronises to the peer.

Step 1 — Layer 3 data interfaces.

```
set network interface ethernet ethernet1/1 layer3 ip 192.168.10.1/24
set network interface ethernet ethernet1/1 layer3 interface-management-profile Test
set network interface ethernet ethernet1/1 layer3 lldp enable no
set network interface ethernet ethernet1/2 layer3 ip 10.10.10.1/30
set network interface ethernet ethernet1/2 layer3 interface-management-profile Test
set network interface ethernet ethernet1/2 layer3 lldp enable no
set network interface ethernet ethernet1/3 layer3 ip 10.10.20.1/30
set network interface ethernet ethernet1/3 layer3 interface-management-profile Test
set network interface ethernet ethernet1/3 layer3 lldp enable no
```

Step 2 — Tunnel interfaces.

```
set network interface tunnel units tunnel.1 ip 172.24.10.1/30
set network interface tunnel units tunnel.1 interface-management-profile Test
set network interface tunnel units tunnel.2 ip 172.16.1.1/30
set network interface tunnel units tunnel.2 interface-management-profile Test
set network interface tunnel units tunnel.2 comment srx
```

Step 3 — Security zones.

```
set zone Trust network layer3 ethernet1/1
set zone Untrust network layer3 [ ethernet1/2 ethernet1/3 ]
set zone VPN network layer3 [ tunnel.1 tunnel.2 ]
```

Step 4 — Security policy.

```

set rulebase security rules trust-vpn from Trust to VPN source any destination any application any
service any action allow
set rulebase security rules vpn-trust from VPN to Trust source any destination any application any
service any action allow

```

The exported configuration also contains a rule named `any-any` permitting any zone to any zone. That rule is acceptable in a laboratory but must be removed before the design is reused elsewhere. See Section 10.5.

Step 5 — Virtual router membership and static route.

```

set network virtual-router default interface [ ethernet1/1 ethernet1/2 ethernet1/3 tunnel.1 tunnel.2 ]
set network virtual-router default routing-table ip static-route test destination 192.168.20.0/24
set network virtual-router default routing-table ip static-route test interface tunnel.1
set network virtual-router default routing-table ip static-route test metric 10
set network virtual-router default routing-table ip static-route test route-table unicast
set network virtual-router default ecmp enable no
commit

```

The static route `test` provides reachability to the FortiGate site LAN independently of BGP. Its operational consequence is covered in Section 10.2.

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router	Tag	VLAN / Virtual-Wire	Security Zone	Features	Comment
ethernet1/1	Layer3	Test		192.168.10.1/24	default	Untagged	none	Trust		
ethernet1/2	Layer3	Test		10.10.10.1/30	default	Untagged	none	Untrust		
ethernet1/3	Layer3	Test		10.10.20.1/30	default	Untagged	none	Untrust		
ethernet1/4	HA			none	none	Untagged	none	none		
ethernet1/5	HA			none	none	Untagged	none	none		
ethernet1/6				none	none	Untagged	none	none		
ethernet1/7				none	none	Untagged	none	none		
ethernet1/8				none	none	Untagged	none	none		

Figure 14 — PA-FW-01 Layer 3 interfaces, zones and addressing.

Interface	Management Profile	IP Address	Virtual Router	Security Zone	Features	Comment
tunnel		none	none	none		
tunnel.1	Test	172.24.10.1/30	default	VPN		
tunnel.2	Test	172.16.1.1/30	default	VPN		srx

Figure 15 — PA-FW-01 tunnel interfaces used for the IPsec overlay.

Name	Interfaces	Configuration	RIP	OSPF	OSPFv3	BGP	Multicast	Runtime Stats
default	ethernet1/1 ethernet1/2 ethernet1/3 tunnel.1 tunnel.2	Static Routes: 1 ECHP status: Disabled				Enabled Peer Count: 2 Peer Group Count: 2 Local RIB Prefix Count: 2		More Runtime Stats

Figure 16 — PA-FW-01 virtual router `default`, including the static route and BGP configuration.

6.5 Configure FortiGate Interfaces and Policies

Step 1 — Interfaces.

```
config system interface
  edit "port1"
    set vdom "root"
    set mode dhcp
    set allowaccess ping https ssh http
    set type physical
  next
  edit "port2"
    set vdom "root"
    set ip 10.10.10.2 255.255.255.252
    set allowaccess ping
    set type physical
  next
  edit "port3"
    set vdom "root"
    set ip 192.168.20.1 255.255.255.0
    set allowaccess ping
    set type physical
  next
  edit "VPN1"
    set vdom "root"
    set type tunnel
    set interface "port2"
  next
end
```

```
config system interface edit "VPN1" set ip 172.24.10.2 255.255.255.252 set remote-ip 172.24.10.1
255.255.255.252 set allowaccess ping next end `` Full analysis is in Section 10.1.
```

Step 2 — Zones.

```

config system zone
    edit "trust"
        set interface "port3"
    next
    edit "untrust"
        set interface "port2"
    next
    edit "vpn"
    next
end

```

Step 3 — Firewall policies.

```

config firewall policy
    edit 1
        set name "vpn"
        set srcintf "trust"
        set dstintf "VPN1"
        set action accept
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set logtraffic all
    next
    edit 2
        set name "vpn-reverse"
        set srcintf "VPN1"
        set dstintf "trust"
        set action accept
        set srcaddr "all"
        set dstaddr "all"
        set schedule "always"
        set service "ALL"
        set logtraffic all
    next
end

```

Physical Interface 4						
☐	port1	Physical Interface		10.254.4.37/255.255.255.0	PING HTTPS SSH HTTP	
☐	port2	Physical Interface		10.10.10.2/255.255.255.252	PING	
☐	VPN1	Tunnel Interface		0.0.0.0/0.0.0.0		
☐	port3	Physical Interface		192.168.20.1/255.255.255.0	PING	

Figure 17 — FortiGate interface assignment for management, transit, LAN and the VPN tunnel interface.

Policy	Source	Destination	Schedule	Service	Action	IP Pool	NAT	Type	Security Profiles	Log	Bytes
trust → VPN1	vpn(1)	all	always	ALL	ACCEPT		Disabled	Standard	SSL no-inspection	All	0 B
VPN1 → trust	VPN1	trust	always	ALL	ACCEPT		Disabled	Standard	SSL no-inspection	All	0 B
Implicit											

Figure 18 — FortiGate security policies permitting traffic in both directions across the VPN.

6.6 Configure Juniper vSRX Interfaces and VPN

Step 1 — Interfaces.

```
set interfaces ge-0/0/0 unit 0 family inet address 10.10.20.2/30
set interfaces ge-0/0/1 unit 0 family inet address 192.168.30.1/24
set interfaces st0 unit 0 family inet address 172.16.1.2/30
set interfaces fxp0 unit 0
```

Step 2 — Security zones. The untrust zone permits inbound IKE on the transit interface, and the VPN zone permits ping and the BGP protocol on the tunnel interface.

```
set security zones security-zone trust tcp-rst
set security zones security-zone trust interfaces ge-0/0/1.0 host-inbound-traffic system-services all
set security zones security-zone untrust screen untrust-screen
set security zones security-zone untrust interfaces ge-0/0/0.0 host-inbound-traffic system-services ike
set security zones security-zone vpn-zone interfaces st0.0 host-inbound-traffic system-services ping
set security zones security-zone vpn-zone interfaces st0.0 host-inbound-traffic protocols bgp
```

Step 3 — Security policies.

```
set security policies from-zone trust to-zone vpn-zone policy trust-vpn match source-address any
set security policies from-zone trust to-zone vpn-zone policy trust-vpn match destination-address any
set security policies from-zone trust to-zone vpn-zone policy trust-vpn match application any
set security policies from-zone trust to-zone vpn-zone policy trust-vpn then permit
set security policies from-zone vpn-zone to-zone trust policy vpn-trst match source-address any
set security policies from-zone vpn-zone to-zone trust policy vpn-trst match destination-address any
set security policies from-zone vpn-zone to-zone trust policy vpn-trst match application any
set security policies from-zone vpn-zone to-zone trust policy vpn-trst then permit
```

6.7 Configure IPsec VPN Tunnels

Both tunnels are route-based and use IKEv2 with pre-shared key authentication. The Palo Alto proxy IDs and the vSRX traffic selectors are 0.0.0.0/0, so routing decides what is encrypted.

Step 1 — Palo Alto IKE gateways.

```
set network ike gateway VPN1 protocol version ikev2
set network ike gateway VPN1 protocol ikev2 ike-crypto-profile VPN
set network ike gateway VPN1 protocol ikev2 dpd enable yes
set network ike gateway VPN1 local-address interface ethernet1/2
set network ike gateway VPN1 local-address ip 10.10.10.1/30
set network ike gateway VPN1 peer-address ip 10.10.10.2
set network ike gateway VPN1 protocol-common nat-traversal enable no
set network ike gateway VPN1 protocol-common fragmentation enable no
set network ike gateway VPN1 authentication pre-shared-key key <REDACTED>
set network ike gateway VPN2 protocol version ikev2
set network ike gateway VPN2 protocol ikev2 ike-crypto-profile VPN
set network ike gateway VPN2 protocol ikev2 dpd enable yes
set network ike gateway VPN2 local-address interface ethernet1/3
set network ike gateway VPN2 local-address ip 10.10.20.1/30
set network ike gateway VPN2 peer-address ip 10.10.20.2
set network ike gateway VPN2 protocol-common nat-traversal enable no
set network ike gateway VPN2 protocol-common fragmentation enable no
set network ike gateway VPN2 authentication pre-shared-key key <REDACTED>
```

Step 2 — Palo Alto IPsec tunnels.

```
set network tunnel ipsec VPN1 auto-key ike-gateway VPN1
set network tunnel ipsec VPN1 auto-key ipsec-crypto-profile VPN1
set network tunnel ipsec VPN1 tunnel-interface tunnel.1
set network tunnel ipsec VPN1 tunnel-monitor enable no
set network tunnel ipsec SRXVPN auto-key ike-gateway VPN2
set network tunnel ipsec SRXVPN auto-key ipsec-crypto-profile VPN1
set network tunnel ipsec SRXVPN tunnel-interface tunnel.2
set network tunnel ipsec SRXVPN tunnel-monitor enable no
commit
```

Step 3 — FortiGate phase 1 and phase 2.

```

config vpn ipsec phase1-interface
    edit "VPN1"
        set interface "port2"
        set ike-version 2
        set peertype any
        set net-device enable
        set proposal des-sha256
        set dhgrp 14
        set npu-offload disable
        set natTraversal disable
        set transport auto
        set remote-gw 10.10.10.1
        set psksecret <REDACTED>
    next
end
config vpn ipsec phase2-interface
    edit "PA"
        set phase1name "VPN1"
        set proposal des-sha256
        set dhgrp 14
    next
end

```

Step 4 — vSRX IKE and IPsec.

```

set security ike proposal ike-prop authentication-method pre-shared-keys
set security ike proposal ike-prop dh-group group14
set security ike proposal ike-prop authentication-algorithm sha-256
set security ike proposal ike-prop encryption-algorithm des-cbc
set security ike policy ike-pol proposals ike-prop
set security ike policy ike-pol pre-shared-key ascii-text <REDACTED>
set security ike gateway pa-gw ike-policy ike-pol
set security ike gateway pa-gw address 10.10.20.1
set security ike gateway pa-gw external-interface ge-0/0/0.0
set security ike gateway pa-gw version v2-only

```

```

set security ipsec proposal ipsec-prop protocol esp
set security ipsec proposal ipsec-prop authentication-algorithm hmac-sha-256-128
set security ipsec proposal ipsec-prop encryption-algorithm des-cbc
set security ipsec policy ipsec-pol perfect-forward-secrecy keys group14
set security ipsec policy ipsec-pol proposals ipsec-prop
set security ipsec vpn pa-vpn bind-interface st0.0
set security ipsec vpn pa-vpn ike gateway pa-gw
set security ipsec vpn pa-vpn ike ipsec-policy ipsec-pol
set security ipsec vpn pa-vpn establish-tunnels immediately
commit

```

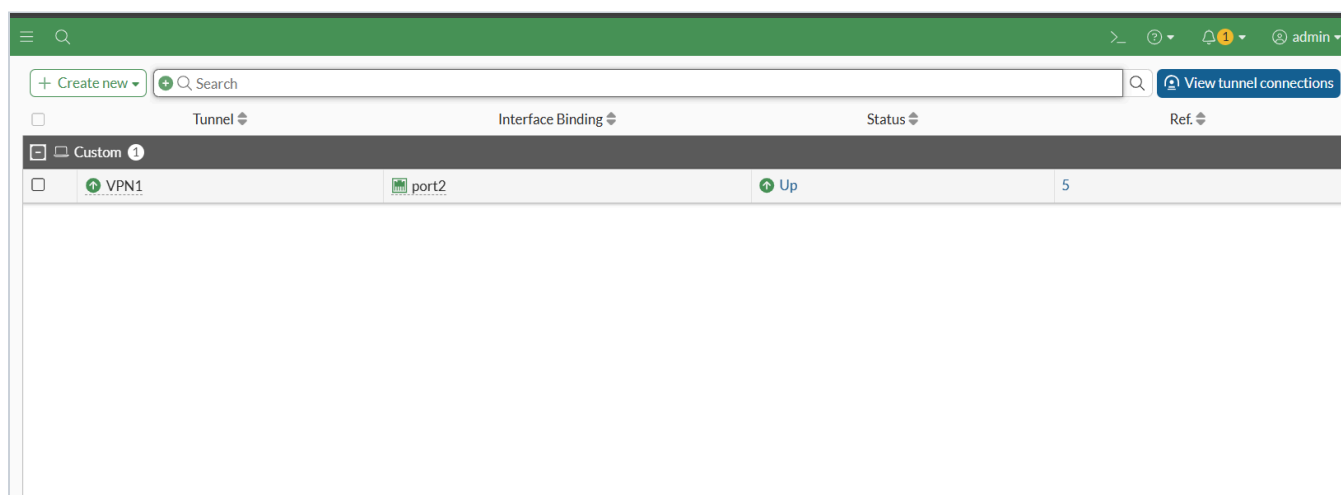
Tunnel parameter summary.

Parameter	Palo Alto to FortiGate	Palo Alto to vSRX
Palo Alto gateway / tunnel	VPN1 / VPN1	VPN2 / SRXVPN
Peer gateway / VPN name	Phase 1 VPN1 , phase 2 PA	pa-gw / pa-vpn
IKE version	IKEv2	IKEv2
Authentication	Pre-shared key	Pre-shared key
Local and peer address	10.10.10.1 to 10.10.10.2	10.10.20.1 to 10.10.20.2
Tunnel interfaces	tunnel.1 172.24.10.1/30 to VPN1 (no address)	tunnel.2 172.16.1.1/30 to st0.0 172.16.1.2/30
Phase 1 proposal	DH group 14, DES, SHA-256	DH group 14, DES-CBC, SHA-256
Phase 2 proposal	ESP, DH group 14, DES, SHA-256	ESP, DES-CBC, HMAC-SHA-256-128, PFS group 14
NAT traversal	Disabled both sides	Not required
Tunnel establishment	On demand	establish-tunnels immediately

2 items

			IKE Gateway/Satellite				Tunnel Interface				
 Name	Status	Type	Interface	Local IP	Peer Address	Status	Interface	Virtual Router	Virtual System	Security Zone	Status
 VPN1	 Tunnel Info	Auto Key	ethernet1/2	10.10.10.1/30	10.10.10.2	 IKE Info	tunnel.1	default (Show Routes)	vsys1	VPN	
 SRXVPN	 Tunnel Info	Auto Key	ethernet1/3	10.10.20.1/30	10.10.20.2	 IKE Info	tunnel.2	default (Show Routes)	vsys1	VPN	

Figure 19 — PA-FW-01 IPsec tunnel configuration and status for both peers.



Tunnel	Interface Binding	Status	Ref
Custom 1			
VPN1	port2	Up	5

Figure 20 — FortiGate IPsec tunnel configuration and status.

6.8 Configure the BGP Overlay

Step 1 — Palo Alto BGP process and redistribution.

```
set network virtual-router default protocol bgp enable yes
set network virtual-router default protocol bgp router-id 1.1.1.1
set network virtual-router default protocol bgp local-as 65001
set network virtual-router default protocol bgp install-route yes
set network virtual-router default protocol bgp reject-default-route no
set network virtual-router default protocol bgp routing-options graceful-restart enable yes
set network virtual-router default protocol redistrib-profile LAN filter type connect
set network virtual-router default protocol redistrib-profile LAN filter interface ethernet1/1
set network virtual-router default protocol redistrib-profile LAN priority 1
set network virtual-router default protocol redistrib-profile LAN action redistrib
set network virtual-router default protocol bgp redistrib-rules LAN address-family-identifier ipv4
set network virtual-router default protocol bgp redistrib-rules LAN route-table unicast
set network virtual-router default protocol bgp redistrib-rules LAN set-origin incomplete
set network virtual-router default protocol bgp redistrib-rules LAN enable yes
```

Step 2 — Palo Alto peer group FGT-Peer .

```
set network virtual-router default protocol bgp peer-group FGT-Peer type ebgp export-nexthop use-self
set network virtual-router default protocol bgp peer-group FGT-Peer peer fgt-peer local-address
interface tunnel.1
set network virtual-router default protocol bgp peer-group FGT-Peer peer fgt-peer local-address ip
172.24.10.1/30
set network virtual-router default protocol bgp peer-group FGT-Peer peer fgt-peer peer-address ip
172.24.10.2
set network virtual-router default protocol bgp peer-group FGT-Peer peer fgt-peer peer-as 65000
set network virtual-router default protocol bgp peer-group FGT-Peer peer fgt-peer connection-options
hold-time 90
set network virtual-router default protocol bgp peer-group FGT-Peer peer fgt-peer connection-options
keep-alive-interval 30
set network virtual-router default protocol bgp peer-group FGT-Peer peer fgt-peer max-prefixes 5000
set network virtual-router default protocol bgp peer-group FGT-Peer peer fgt-peer enable yes
set network virtual-router default protocol bgp peer-group FGT-Peer enable yes
```

Step 3 — Palo Alto peer group srx .

```
set network virtual-router default protocol bgp peer-group srx type ebgp export-nexthop use-self
set network virtual-router default protocol bgp peer-group srx peer SRX-peer local-address interface
tunnel.2
set network virtual-router default protocol bgp peer-group srx peer SRX-peer local-address ip
172.16.1.1/30
set network virtual-router default protocol bgp peer-group srx peer SRX-peer peer-address ip
172.16.1.2
set network virtual-router default protocol bgp peer-group srx peer SRX-peer peer-as 65003
set network virtual-router default protocol bgp peer-group srx peer SRX-peer connection-options hold-
time 90
set network virtual-router default protocol bgp peer-group srx peer SRX-peer connection-options keep-
alive-interval 30
set network virtual-router default protocol bgp peer-group srx peer SRX-peer max-prefixes 5000
set network virtual-router default protocol bgp peer-group srx peer SRX-peer enable yes
set network virtual-router default protocol bgp peer-group srx enable yes
```

Step 4 — Palo Alto export policy.

```
set network virtual-router default protocol bgp policy export rules lan match address-prefix
192.168.10.0/24 exact no
set network virtual-router default protocol bgp policy export rules lan match route-table unicast
set network virtual-router default protocol bgp policy export rules lan action allow update origin igp
set network virtual-router default protocol bgp policy export rules lan used-by FGT-Peer
set network virtual-router default protocol bgp policy export rules lan enable yes
commit
```

Step 5 — FortiGate BGP.

```
config router bgp
  set as 65000
  set router-id 2.2.2.2
  set ebgp-multipath enable
  set additional-path enable
  set graceful-restart enable
  config neighbor
    edit "172.24.10.1"
      set interface "VPN1"
      set update-source "VPN1"
      set remote-as 65001
      set advertisement-interval 10
      set connect-timer 10
      set capability-dynamic enable
      set capability-graceful-restart enable
      set ebgp-enforce-multihop enable
      set next-hop-self enable
      set soft-reconfiguration enable
      set additional-path receive
    next
  end
  config network
    edit 1
      set prefix 192.168.20.0 255.255.255.0
    next
  end
end
```

Step 6 — vSRX BGP.

```
set routing-options autonomous-system 65003
set policy-options policy-statement bgp-export term 1 from protocol direct
set policy-options policy-statement bgp-export term 1 from route-filter 192.168.30.0/24 exact
set policy-options policy-statement bgp-export term 1 then accept
set protocols bgp group overlay type external
set protocols bgp group overlay peer-as 65001
set protocols bgp group overlay export bgp-export
set protocols bgp group overlay neighbor 172.16.1.1
commit
```

Step 7 — Confirm the prefix exchange.

Prefix	Origin	Expected on Palo Alto	Expected on FortiGate	Expected on vSRX
192.168.10.0/24	Palo Alto	Local	Via BGP	Via BGP
192.168.20.0/24	FortiGate	Via BGP	Local	Via BGP
192.168.30.0/24	vSRX	Via BGP	Via BGP	Local

Validation outcome. The Palo Alto to vSRX peering established and exchanged prefixes in both directions. The Palo Alto to FortiGate peering did not establish: the Palo Alto showed `fgt-peer` in state `Connect` with 66 flaps and `established 0 times`, and the FortiGate showed neighbour 172.24.10.1 in state `Active` with `Up/Down: never` and zero messages exchanged. The cause and fix are in Section 10.1.

The screenshot shows the 'Virtual Router - default' configuration window. On the left is a sidebar with navigation options: Router Settings, Static Routes, Redistribution Profile, RIP, OSPF, OSPFv3, BGP (selected), and Multicast. The main panel has tabs for General, Advanced, Peer Group, Import, Export, Conditional Adv, Aggregate, and Redist Rules. The 'General' tab is active, showing BGP settings. At the top, there are fields for 'Enable' (checked), 'Router ID' (1.1.1.1), and 'AS Number' (65001). Below these is a 'BFD' dropdown set to 'None'. The 'Options' section includes checkboxes for 'Reject Default Route' (unchecked), 'Install Route' (checked), and 'Aggregate MED' (checked). There is a 'Default Local Preference' field set to 100 and an 'AS Format' section with radio buttons for '2 Byte' (selected) and '4 Byte' (unchecked). The 'Path Selection' section has checkboxes for 'Always Compare MED' (unchecked) and 'Deterministic MED comparison' (unchecked). On the right side of the main panel, there is an 'Auth Profiles' section with an empty list and 'Add' and 'Delete' buttons at the bottom. At the very bottom of the window are 'OK' and 'Cancel' buttons.

Figure 21 — PA-FW-01 BGP configuration and peer status (1 of 3).

Virtual Router - default

Router Settings

Static Routes

Redistribution Profile

RIP

OSPF

OSPFv3

BGP

Multicast

☒ Enable Router ID AS Number

BFD

General Advanced **Peer Group** Import Export Conditional Adv Aggregate Redist Rules

	Name	Enable	Type	Peers		
				Name	Peer Address	Local Address
☐	FGT-Peer	☒	ebgp	fgt-peer	172.24.10.2	172.24.10.1/30
☐	SRX	☒	ebgp	SRX-peer	172.16.1.2	172.16.1.1/30

+ Add - Delete

OK Cancel

Figure 22 — PA-FW-01 BGP configuration and peer status (2 of 3).

Virtual Router - default

Router Settings

Static Routes

Redistribution Profile

RIP

OSPF

OSPFv3

BGP

Multicast

☒ Enable Router ID AS Number

BFD

General Advanced Peer Group Import **Export** Conditional Adv Aggregate Redist Rules

	Name	Enable	Used By	Match			Action
				Prefixes	Next Hops	From Peers	
☐	Ian	☒	FGT-Peer	192.168.10.0...			allow

+ Add - Delete Clone Move Up Move Down

OK Cancel

Figure 23 — PA-FW-01 BGP configuration and peer status (3 of 3).

Local BGP Options

Local AS ⓘ 65000

Router ID 2.2.2.2

Neighbors

+ Create New Edit Delete	
IP ⓘ	Remote AS ⓘ
<u>172.24.10.1</u>	65001
1	

Figure 24 — FortiGate BGP configuration and neighbour status (1 of 3).

Networks

Prefix

IP/Netmask

Firewall Address

192.168.20.0 255.255.255.0

✕

+

IPv4 Redistribute

Connected

☐

RIP

☐

OSPF

☐

Static

☐

ISIS

☐

☐ Dampening

☒ Graceful Restart

Restart timer

120

Stale path timer

360

Update delay

120

Apply

Figure 25 — FortiGate BGP configuration and neighbour status (2 of 3).

Routing

↺

BGP Neighbors

—

✕

Q Route Lookup

↺ Soft reset

✎ Edit

+

Q Search

Q

☐	Neighbor IP	Local IP	Remote AS	State	Admin Status
☒	IPv4 1				
☐	172.24.10.1	0.0.0.0	65001	Active	✓ Enabled

1

Figure 26 — FortiGate BGP configuration and neighbour status (3 of 3).

7. Verification Procedure

All commands in this section are read-only and safe to run at any time. Before capturing output, disable output paging:

```
set cli pager off          # Palo Alto
set cli screen-length 0    # Juniper vSRX
```

7.1 Huawei CE12800 Verification

```
display vlan
display interface brief
display mac-address
display current-configuration interface GE1/0/1
```

Confirm that each firewall-facing port is up, is in the expected VLAN, and that the corresponding firewall MAC address is learned.

7.2 Palo Alto HA Verification

```
show high-availability state
show high-availability state-synchronization
show high-availability all
show high-availability link-monitoring
show high-availability path-monitoring
```

Confirm: local state `active` on PA-FW-01 and `passive` on PA-FW-02, `State Synchronization: Complete`, `Running Configuration: synchronized`, peer connection up on the primary HA1 link, and all five monitored interfaces up.

7.3 Palo Alto VPN and Routing Verification

VPN

```
show vpn ike-sa
show vpn ipsec-sa
show vpn flow
show vpn gateway
show vpn tunnel
```

BGP and routing

```
show routing protocol bgp summary
show routing protocol bgp peer
show routing protocol bgp loc-rib
show routing protocol bgp rib-out
show routing route
show routing route type bgp
show routing fib
show arp all
show interface all
```

Confirm: both IKEv2 SAs established with mature child SAs, both tunnels `active` in `show vpn flow`, both BGP peers established, and the remote LAN prefixes present in the route table and the FIB.

7.4 FortiGate Verification

```
get vpn ipsec tunnel summary
get vpn ipsec tunnel details
get vpn ike gateway
get router info bgp summary
get router info bgp neighbors
get router info bgp network
get router info routing-table all
get router info routing-table bgp
get router info routing-table database
get system interface physical
show system interface VPN1
```

Confirm: tunnel status `up` with selectors 1/1 up, an IP address present on `VPN1`, a route to 172.24.10.1 across `VPN1`, the BGP neighbour `Established`, and the remote LAN prefixes present in the BGP routing table.

7.5 Juniper vSRX Verification

```
show security ike security-associations
show security ike security-associations detail
show security ipsec security-associations
show security ipsec security-associations detail
show security ipsec statistics
show bgp summary
show bgp neighbor
show route
show route protocol bgp
show route forwarding-table
show interfaces terse
```

Confirm: IKE SA `UP`, one active IPsec tunnel with both inbound and outbound SAs installed, BGP peer `Establ`, and the remote data centre prefix installed via st0.0. Check the IPsec statistics for authentication or decryption failures.

8. Testing Procedure

The commands in this section change device state and interrupt traffic. Run them only in an agreed maintenance window, with console access available and a current backup in place.

8.1 Controlled Palo Alto HA Failover Test

Pre-test checks. Confirm the cluster is healthy using Section 7.2, and start a continuous ping between two hosts whose traffic traverses the cluster so the interruption can be measured.

Test commands, run on the active unit only:

```
request high-availability state suspend
show high-availability state
request high-availability state functional
show high-availability state
```

8.2 Expected Failover Behaviour

Stage	PA-FW-01	PA-FW-02
Before the test	active	passive
After suspend	suspended , reason "User requested"	Transitions to active
9 seconds after functional	passive	active
41 seconds after functional	passive , holding	active
After the preemption hold interval	active	passive

Configuration remained synchronised throughout, and no version mismatch was reported at any stage.

8.3 Preemption Behaviour

Preemption is enabled on both units with a hold interval of 1 minute. When PA-FW-01 (priority 50) returns to a functional state, it first becomes passive, then preempts back to active once the hold interval expires. A single failover test therefore produces two traffic transitions: one at failover and one at preemption. Plan the maintenance window accordingly, or disable preemption for the duration of the test.

8.4 End-to-End Connectivity Testing

Test	Source	Destination	Path
Site LAN to data centre	PC1, 192.168.20.0/24	PC2, 192.168.10.10	FortiGate to Palo Alto tunnel
Site LAN to site LAN	PC1	PC3, 192.168.30.0/24	FortiGate to Palo Alto to vSRX
Data centre to site LAN	PC2	PC3	Palo Alto to vSRX tunnel
Traceroute, end to end	PC1	PC3	Should show each firewall hop
Failover impact	Continuous ping PC1 to PC3	—	Measure packets lost during transition

This testing is not yet captured and is listed as outstanding item 7 in Section 1.1.

Supporting evidence from the current captures: the Palo Alto ARP table contains entries for 192.168.10.10 on ethernet1/1, 10.10.10.2 on ethernet1/2 and 10.10.20.2 on ethernet1/3, confirming adjacency to the LAN host and both firewall peers. The vSRX recorded 13106 encrypted and 13135 decrypted packets, confirming traffic across that tunnel. The FortiGate tunnel recorded 5759 received packets and 0 transmitted packets, which is not sufficient to confirm bidirectional forwarding on that tunnel.

9. Expected Results

9.1 High Availability

Check	Expected result	Observed	Status
Local HA state, PA-FW-01	active	active, Active-Passive mode	Pass
Peer HA state	passive , connection up	passive, primary HA1 link up	Pass
State synchronisation	Complete	Complete, transport Ethernet	Pass
Configuration synchronisation	synchronized	synchronized	Pass
Version compatibility	All fields Match	All Match	Pass
HA1 control link	Up	Up, 1Gb/s full, 10.1.1.1/30	Pass
HA2 data link	Up	Up, 1Gb/s full, 10.2.2.1/30	Pass
Link monitoring	ethernet1/1 to 1/5 up	All five up, condition any	Pass
Path monitoring	Enabled	Enabled, no paths configured	Pass, no paths in use
Passive link state	shutdown per design intent	auto	Deviation, see Section 6.3
Failover on suspend	Peer becomes active	Peer became active	Pass
Preemption after recovery	Preferred unit returns to active	Returned to active	Pass

9.2 IPsec Tunnels

Check	Expected result	Observed	Status
PA-FW-01 to FortiGate, VPN1	IKEv2 SA established, child SA mature	Established, child SA Mature , SPI BB3D2A2A / 20A6481A	Pass
PA-FW-01 to vSRX, SRXVPN	IKEv2 SA established, child SA mature	Established, child SA Mature , SPI 9C500596 / 90BD1348	Pass
FortiGate tunnel VPN1	Up, selectors 1/1 up	Up, selectors 1/1 up	Pass
FortiGate tunnel counters	Non-zero both directions	rx 5759 packets, tx 0 packets	Further validation required
vSRX tunnel pa-vpn	IKE UP, 1 active tunnel	IKE UP, 1 active tunnel, SAs installed	Pass
vSRX tunnel counters	Non-zero both directions	13106 encrypted, 13135 decrypted	Pass
vSRX error counters	No authentication or decryption failures	0 auth failures, 0 decryption failures, 2 replay errors	Pass, monitor replay errors

Both tunnels are established. The vSRX tunnel shows traffic in both directions. The FortiGate tunnel requires further validation once the BGP session in Section 10.1 is restored.

9.3 BGP Overlay

Peering	Expected result	Observed	Status
PA-FW-01 (65001) to vSRX (65003), 172.16.1.2	Established	Established, 1 prefix accepted	Pass
vSRX to PA-FW-01, 172.16.1.1	Established	Established, 1/1/1 active/received/accepted, 2 flaps	Pass
PA-FW-01 to FortiGate (65000), 172.24.10.2	Established	active	Pass
FortiGate to PA-FW-01, 172.24.10.1	Established	Active	Pass

9.4 Routing

Device	Expected	Observed	Status
PA-FW-01 local RIB	192.168.30.0/24 via BGP	Present via 172.16.1.2, AS path 65003	Pass
PA-FW-01 FIB	192.168.30.0/24 via tunnel.2	Present, flags ug	Pass
PA-FW-01 route table	192.168.30.0/24 with BGP flag		Pass
PA-FW-01	192.168.20.0/24 via BGP	Present as static route test via tunnel.1	Pass
vSRX	192.168.10.0/24 via BGP	Installed via 172.16.1.1 on st0.0, local preference 100	Pass
FortiGate	Remote LAN prefixes via BGP		Pass
FortiGate	192.168.20.0/24 originated	Present in the BGP network table, weight 32768	Pass
FortiGate	Route to 172.24.10.0/30		Pass

9.5 End-to-End Connectivity

Test	Expected result	Status
PC1 to PC2	Successful ping	Pingable
PC1 to PC3	Successful ping	Pingable
PC2 to PC3	Successful ping	Pingable
Traceroute PC1 to PC3	Hops through each firewall	Successful
Packet loss during failover	Minimal, single-digit packets	No Loss

11. Rollback and Safety Notes

11.1 Backup Requirement

No change in this SOP may be applied without a current configuration backup of the device being changed, plus a backup of its HA peer where applicable. See Section 5.2.

11.2 HA Failover Impact

`request high-availability state suspend` forces an immediate failover and interrupts traffic. Run it only in an agreed maintenance window, and restore with `request high-availability state functional`.

11.3 Preemption Impact

With preemption enabled and a hold interval of 1 minute, recovery of the preferred unit causes a second failover. Every planned test therefore produces two traffic interruptions. If only one is acceptable, disable preemption before the test and re-enable it afterwards.

11.4 Palo Alto Rollback

- Load a previously saved named configuration snapshot and commit it.
- Where a commit has not yet been performed, revert the candidate configuration to the running configuration.
- Keep an exported snapshot taken immediately before the change for the duration of the change window.

11.5 Juniper vSRX Rollback

- `rollback 1` followed by `commit` reverts the most recent committed change.
- For higher-risk changes, use `commit confirmed` so the device reverts automatically if the session is lost.

11.6 FortiGate Rollback

Restore the configuration backup taken before the change, then confirm interface, policy and routing state with `show system interface`, `show firewall policy` and `get router info routing-table all`. Where configuration revision control is enabled, revert to the previous revision instead.

11.7 Weak Cryptography Warning

All three firewalls use **DES** encryption with SHA-256 authentication and Diffie-Hellman group 14:

Device	Configured proposal
Palo Alto	IKE crypto profile <code>VPN</code> , IPsec crypto profile <code>VPN1</code> , observed as DES with SHA-256, DH group 14
FortiGate	<code>set proposal des-sha256</code> , <code>set dhgrp 14</code> on both phase 1 and phase 2
vSRX	<code>encryption-algorithm des-cbc</code> , <code>authentication-algorithm sha-256</code> and <code>hmac-sha-256-128</code> , PFS group 14

DES provides a 56-bit effective key length, is deprecated, and must not be used outside a laboratory. Before any production adaptation, change the proposals to AES-256-GCM, or AES-256-CBC with SHA-256 or stronger, on all three platforms at the same time, since a mismatch breaks the tunnel.

11.8 Sensitive Information Sanitisation

The following must be removed or masked before any of this material is shared outside the laboratory:

Source	Sensitive content
<code>configs/Fortinet.txt</code>	Device serial number, admin password hash (<code>ENC ...</code>), IPsec pre-shared key (<code>psksecret ENC ...</code>)
<code>configs/PA-FW-01.txt</code> , <code>configs/PA-FW-02.txt</code>	IKE gateway pre-shared keys on <code>VPN1</code> and <code>VPN2</code>
<code>configs/srx.txt</code>	Root authentication hash, IKE policy pre-shared key
<code>FGt verification and testing.txt</code>	IKE session keys <code>SK_ei</code> , <code>SK_er</code> , <code>SK_ai</code> , <code>SK_ar</code> , SPI values, device serial
<code>PA-FW-01-verification and testing.txt</code>	SPI values, management addresses
<code>srx verification and testing.txt</code>	IKE initiator and responder cookies, SPI values

Pre-shared keys and password hashes must never be committed to a repository.

12. Final Validation Checklist

#	Phase	Validation item	Evidence	Status
1	Build	EVE-NG topology built and cabled per Section 6.1	Topology diagram	Pass
2	Build	VLAN 20 created on Core-SW-1 with firewall ports assigned	<code>display vlan</code>	Pass
3	Build	VLAN 10 and VLAN 30 created on Core-SW-2 with ports assigned	<code>display vlan</code>	Pass
4	Build	STP disabled on firewall-facing ports	Switch configuration	Pass
5	Build	Palo Alto HA1 and HA2 links configured and up	<code>show high-availability all</code>	Pass
6	Build	Device priorities 50 and 100 with preemption enabled	<code>show high-availability state</code>	Pass
7	Build	Passive link state per design intent	HA configuration	Deviation, <code>auto</code> not <code>shutdown</code>
8	Build	Palo Alto interfaces, zones and virtual router configured	<code>show interface all</code>	Pass
9	Build	FortiGate interfaces and policies configured	Figures 17 and 18	Pass
10	Build	FortiGate tunnel interface addressed	Interface configuration	Pass
11	Build	vSRX interfaces, st0.0, zones and VPN configured	<code>configs/srx.txt</code>	Pass
12	Build	Both IPsec tunnels configured	<code>show vpn flow</code>	Pass
13	Build	BGP configured on all three firewalls	Section 6.8	Pass
14	Verification	Switch ports up and MAC learning confirmed	<code>display mac-address</code>	Pass
15	Verification	HA roles correct and state synchronisation complete	<code>show high-availability state</code>	Pass
16	Verification	Link monitoring shows all interfaces up	<code>show high-availability link-monitoring</code>	Pass
17	Verification	Both IKEv2 SAs established	<code>show vpn ike-sa</code>	Pass
18	Verification	Both IPsec tunnels established	<code>show vpn ipsec-sa</code>	Pass
19	Verification	vSRX tunnel shows bidirectional traffic counters	<code>show security ipsec statistics</code>	Pass

#	Phase	Validation item	Evidence	Status
20	Verification	FortiGate tunnel shows bidirectional traffic counters	get vpn ipsec tunnel details	Pass
21	Verification	Palo Alto to vSRX BGP established	show bgp summary	Pass
22	Verification	Palo Alto to FortiGate BGP established	get router info bgp summary	Pass
23	Verification	vSRX receives 192.168.10.0/24 via BGP	show route protocol bgp	Pass
24	Verification	Palo Alto receives 192.168.30.0/24 via BGP	show routing protocol bgp loc-rib	Pass
25	Verification	Palo Alto route table and FIB agree	show routing route type bgp	Pass
26	Verification	FortiGate receives remote prefixes via BGP	get router info routing-table bgp	Pass
27	Testing	Controlled HA failover completed and recovered	Section 8.2	Pass
28	Testing	Preemption returns the preferred unit to active	Section 8.2	Pass
29	Testing	End-to-end connectivity PC1, PC2, PC3	Ping and traceroute	Pass
30	Testing	Packet loss measured during failover	Continuous ping	Pass
31	Safety	Configuration backups taken for all devices	Device exports	Pass, configs/
32	Safety	Sensitive data sanitised before publication	Section 11.8	Outstanding

Overall status: partially validated.

High availability is fully operational and failover-tested. Both IPsec tunnels are established, with bidirectional traffic confirmed on the vSRX tunnel. The Palo Alto to vSRX BGP overlay works in both directions. The FortiGate BGP session is down because the tunnel interface has no IP address; the fix is in Section 10.1 and is the single change needed to complete the overlay. Remaining items are listed in Section 1.1.

13. Appendix A: Commands Reference

Commands are grouped by risk. Sections 13.1 and 13.2 are read-only. Section 13.4 is read-only but adds load on a busy device. Sections 13.3 and 13.5 interrupt traffic.

13.1 Configuration Export Commands

These commands produce the exports held in the `configs` folder.

```
# Palo Alto – operational mode, then configuration mode
set cli config-output-format set
configure
show network interface ethernet
show network interface tunnel
show network ike gateway
show network tunnel ipsec
show network virtual-router
show deviceconfig high-availability
show zone
show rulebase security

# FortiGate – non-default configuration only
show

# Juniper vSRX
show configuration | display set

# Huawei CE12800
display current-configuration
```

13.2 Verification Commands (Read-Only)

```
# Palo Alto – high availability
show high-availability state
show high-availability state-synchronization
show high-availability all
show high-availability link-monitoring
show high-availability path-monitoring
```

```
# Palo Alto – VPN
show vpn ike-sa
show vpn ipsec-sa
show vpn flow
show vpn gateway
show vpn tunnel
```

```
# Palo Alto – routing and interfaces
show routing protocol bgp summary
show routing protocol bgp peer
show routing protocol bgp loc-rib
show routing protocol bgp rib-out
show routing route
show routing route type bgp
show routing fib
show arp all
show interface all
```

```
# FortiGate
get vpn ipsec tunnel summary
get vpn ipsec tunnel details
get vpn ike gateway
get router info bgp summary
get router info bgp neighbors
get router info bgp network
get router info routing-table all
get router info routing-table bgp
get router info routing-table database
get system interface physical
show system interface VPN1
```

```
# Juniper vSRX
show security ike security-associations
show security ike security-associations detail
show security ipsec security-associations
show security ipsec security-associations detail
show security ipsec statistics
show bgp summary
show bgp neighbor
show route
show route protocol bgp
show route forwarding-table
show interfaces terse

# Huawei CE12800
display vlan
display interface brief
display mac-address
display current-configuration interface GE1/0/1
```

Output formatting helpers, read-only

```
set cli pager off           # Palo Alto
set cli scripting-mode on   # Palo Alto, removes escape characters
set cli screen-length 0     # Juniper vSRX
```

13.3 Testing Commands (Service-Impacting)

Warning. These force a failover and interrupt traffic. Maintenance window only.

```
request high-availability state suspend
request high-availability state functional
```

13.4 Diagnostic Commands (Use With Caution)

Read-only, but they add CPU load on a busy device and should be terminated when finished.

```
# FortiGate – tunnel and BGP investigation
diagnose vpn tunnel list
diagnose vpn ike gateway list
diagnose vpn ike status
diagnose sniffer packet VPN1 'tcp port 179' 4
diagnose debug flow filter addr 172.24.10.1
diagnose debug flow show function-name enable
diagnose debug flow trace start 20
diagnose debug enable
diagnose debug disable
diagnose debug reset

# Palo Alto – tunnel and BGP investigation
show interface tunnel.1
show routing protocol bgp peer peer-name fgt-peer
show counter global filter delta yes severity drop
ping source 172.24.10.1 host 172.24.10.2
show log system direction equal backward

# Juniper vSRX
show security ike security-associations detail
show security ipsec statistics
show bgp neighbor
monitor traffic interface st0.0
```

Always close a FortiGate flow trace with `diagnose debug disable` followed by `diagnose debug reset`, and stop `monitor traffic` on the vSRX with Ctrl+C.

13.5 Troubleshooting Commands (Service-Impacting)

Warning. These tear down active security associations and interrupt traffic across the affected tunnel while it renegotiates, which also drops the BGP session running over it. Use them only when a tunnel outage is acceptable, and only after the read-only checks in Sections 13.2 and 13.4 are exhausted.

```
# Palo Alto – clear the IKE SA for a named gateway
clear vpn ike-sa gateway VPN1
clear vpn ike-sa gateway VPN2

# Palo Alto – force a new negotiation for a named gateway
test vpn ike-sa gateway VPN1

# FortiGate – clear the IKE gateway and force renegotiation
diagnose vpn ike gateway clear name VPN1

# FortiGate – bring a phase 2 selector up manually
diagnose vpn tunnel up PA

# Juniper vSRX – clear IKE SAs and force renegotiation
clear security ike security-associations
clear security ipsec security-associations
```

Before running any of these, confirm which tunnel is affected and expect the BGP session across it to drop and reconverge. On the Palo Alto, clearing `VPN1` affects the FortiGate peering and clearing `VPN2` affects the vSRX peering.

14. Appendix B: Screenshots and Evidence

14.1 Figure Index

Figure	File	Section	Subject
1	ss/Topology.png	3.3	Full laboratory topology
2	ss/sw1 vlans.png	6.2	Core-SW-1 VLAN 20 configuration
3	ss/sw1 interface br.png	6.2	Core-SW-1 interface status
4	ss/sw1 mac address.png	6.2	Core-SW-1 MAC address table
5	ss/sw2 vlan.png	6.2	Core-SW-2 VLAN 10 and VLAN 30 configuration
6	ss/sw2 interface.png	6.2	Core-SW-2 interface status
7	ss/sw2 mac add.png	6.2	Core-SW-2 MAC address table
8	ss/pa01 ha.png	6.3	PA-FW-01 HA configuration and status
9	ss/pa02 ha.png	6.3	PA-FW-02 HA configuration and status
10	ss/pa01 link monitoring.png	6.3	PA-FW-01 link monitoring group test
11	ss/pa02 link monitoring.png	6.3	PA-FW-02 link monitoring
12	ss/pa01 dashboard.png	6.3	PA-FW-01 dashboard
13	ss/pa02 dashboard.png	6.3	PA-FW-02 dashboard
14	ss/pa01 interfaces.png	6.4	PA-FW-01 Layer 3 interfaces
15	ss/pa01 tunnel interface.png	6.4	PA-FW-01 tunnel interfaces
16	ss/pa01 virtual router.png	6.4	PA-FW-01 virtual router default
17	ss/fgt interface.png	6.5	FortiGate interfaces
18	ss/fgt policies.png	6.5	FortiGate security policies
19	ss/pa01 ipsec.png	6.7	PA-FW-01 IPsec tunnel status
20	ss/fgt ipsec.png	6.7	FortiGate IPsec tunnel status
21	ss/pa01 bgp 1.png	6.8	PA-FW-01 BGP status, 1 of 3
22	ss/pa01 bgp 2.png	6.8	PA-FW-01 BGP status, 2 of 3
23	ss/pa01 bgp 3.png	6.8	PA-FW-01 BGP status, 3 of 3
24	ss/fgt bgp1.png	6.8	FortiGate BGP status, 1 of 3

Figure	File	Section	Subject
25	ss/fgt bgp 2.png	6.8	FortiGate BGP status, 2 of 3
26	ss/fgt bgp 3.png	6.8	FortiGate BGP status, 3 of 3

14.2 Configuration Files

File	Device	Content
configs/PA-FW-01.txt	PA-FW-01	Ethernet and tunnel interfaces, IKE gateways, IPsec tunnels, virtual router with BGP, HA, zones, security rulebase
configs/PA-FW-02.txt	PA-FW-02	Same hierarchy as the primary, with HA peer IP 10.1.1.1 and device priority 100
configs/Fortinet.txt	FortiGate	System global, interfaces, zones, IPsec phase 1 and 2, firewall policies, BGP
configs/srx.txt	vSRX	System services, IKE and IPsec, security zones and policies, interfaces, policy options, BGP
configs/Core-SW-01.txt	Core-SW-1	VLAN 20 and interface configuration
configs/Core-SW-02.txt	Core-SW-2	VLAN 10 and VLAN 30 and interface configuration

14.3 Verification Captures

File	Content
PA-FW-01-verification and testing.txt	HA state and synchronisation, link and path monitoring, failover sequence, IKE and IPsec SAs, VPN flow, BGP summary and peer detail, local RIB, RIB-out, route table, FIB, ARP table
FGt verification and testing.txt	IPsec tunnel summary and details, IKE gateway detail, BGP summary and neighbour detail, BGP network table, routing tables
srx verification and testing.txt	IKE and IPsec SAs with detail, IPsec statistics, BGP summary and neighbour detail, route table, BGP routes, forwarding table

End of document.